

SRE CASE STUDY

AZURE ARC HYBRID CLOUD LAB

Parts 1-3: Kubernetes, Azure Arc, Terraform, CI/CD,
Ingress TLS, HPA, NetworkPolicy, RBAC

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Date: May 2026 (Parts 1-3)

GitHub: github.com/buth11/sre-homelab-azure-arc

Stack: Kubernetes | Azure Arc | Terraform | KQL | Azure DevOps | Traefik | HPA | NetworkPolicy
| RBAC

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Chapter 1: Context and Architecture

This project demonstrates building a production-grade hybrid environment on GMKtec EVO-X2 (128 GB RAM) connecting Kubernetes with Microsoft Azure through Azure Arc. Three parts of hands-on practice with documentation of every step and 10 documented incidents.

Layer	Technology	Purpose
Kubernetes	K3s v1.35.4+k3s1	3-node cluster
Ingress	Traefik + TLS	HTTPS domain routing
Autoscaling	HPA (autoscaling/v2)	CPU-based scaling 2-8 replicas
Network Security	NetworkPolicy	Intra-cluster firewall
Access Control	RBAC + ServiceAccount	Least privilege
Cloud Bridge	Azure Arc	Hybrid control plane
Monitoring	Azure Monitor + Prometheus	Observability
IaC	Terraform + Remote State	Infrastructure as Code
CI/CD	Azure DevOps	Pipeline with approval gate

Chapter 2: On-Premise Infrastructure

Node	IP	CPU	RAM
k3s-master (Control Plane)	192.168.122.10	4 vCPU	8 GB
k3s-worker1	192.168.122.11	4 vCPU	16 GB
k3s-worker2	192.168.122.12	4 vCPU	16 GB

Chapter 3: FinOps

The screenshot displays the Microsoft Azure Cost Management dashboard for user Bartosz Suszko. The interface is in Polish. The left sidebar contains navigation links for various Azure services. The main content area shows the 'Budżety' (Budgets) section. A table lists the budget details for 'SRE-Lab-Budget'.

Nazwa	Zakres	Resetuj okres	Data utworzenia	Data wygaśnięcia	Budżet	Prognostowane	Oszacowane wy...	Postęp
SRE-Lab-Budget	55df28b8-39ed-58b...	Monthly	1.05.2026	30.04.2028	5,00 USD		\$0.00	0.00%

Screenshot 1: Azure Cost Management — 5 USD/month budget, cost: 0.00 USD.

Chapter 4: Azure Arc

The screenshot displays the Microsoft Azure portal interface. On the left, the navigation pane shows the 'Klastry Kubernetes' (Kubernetes Clusters) section under 'Podnieś poziom planu' (Raise the plan). The main content area shows the 'K3s-Homelab' cluster details. The cluster is connected and shows the following information:

- Podstawowe elementy** (Basic elements):
 - Subskrypcja (przenieś): Czas ostatniego połączenia: 2026-05-21T10:40:48.964Z
 - Identyfikator subskrypcji: b023e8b8-4446-4d4f-9fcf-d20a6c1fa093
 - Grupa zasobów (przenieś): SRE-Lab-RG
 - Stan: Connected
 - Ustawienia: Lokalizacja: West Europe, Menedżer floty: Kliknij tutaj, aby przypisać
- Właściwości** (Properties):
 - Wersja agenta: 1.33.0
 - Czas wygaśnięcia certyfikatu tożsamości zarządzanej: August 19, 2026 at 08:20 AM GMT+2
 - Wersja platformy Kubernetes: 1.35.4+k3s1
 - Łączna liczba węzłów: 3
 - Łączna liczba rdzeni: 12

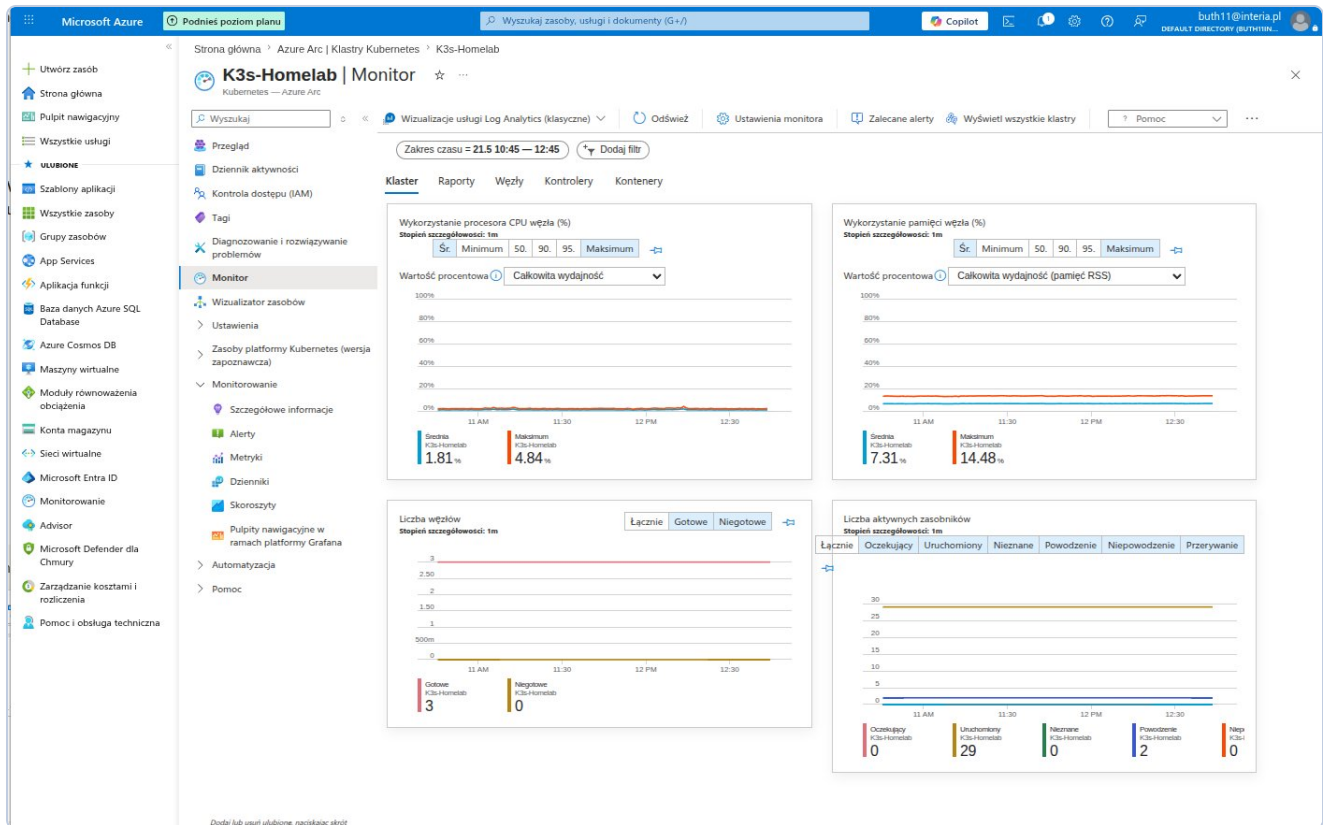
The bottom of the screen shows the cluster's status as 'Connected' and the version 'v1.35.4+k3s1'.

Screenshot 2: Azure Arc — K3s-Homelab Connected, v1.35.4+k3s1, 3 nodes, 12 cores.

Chapter 5: Observability

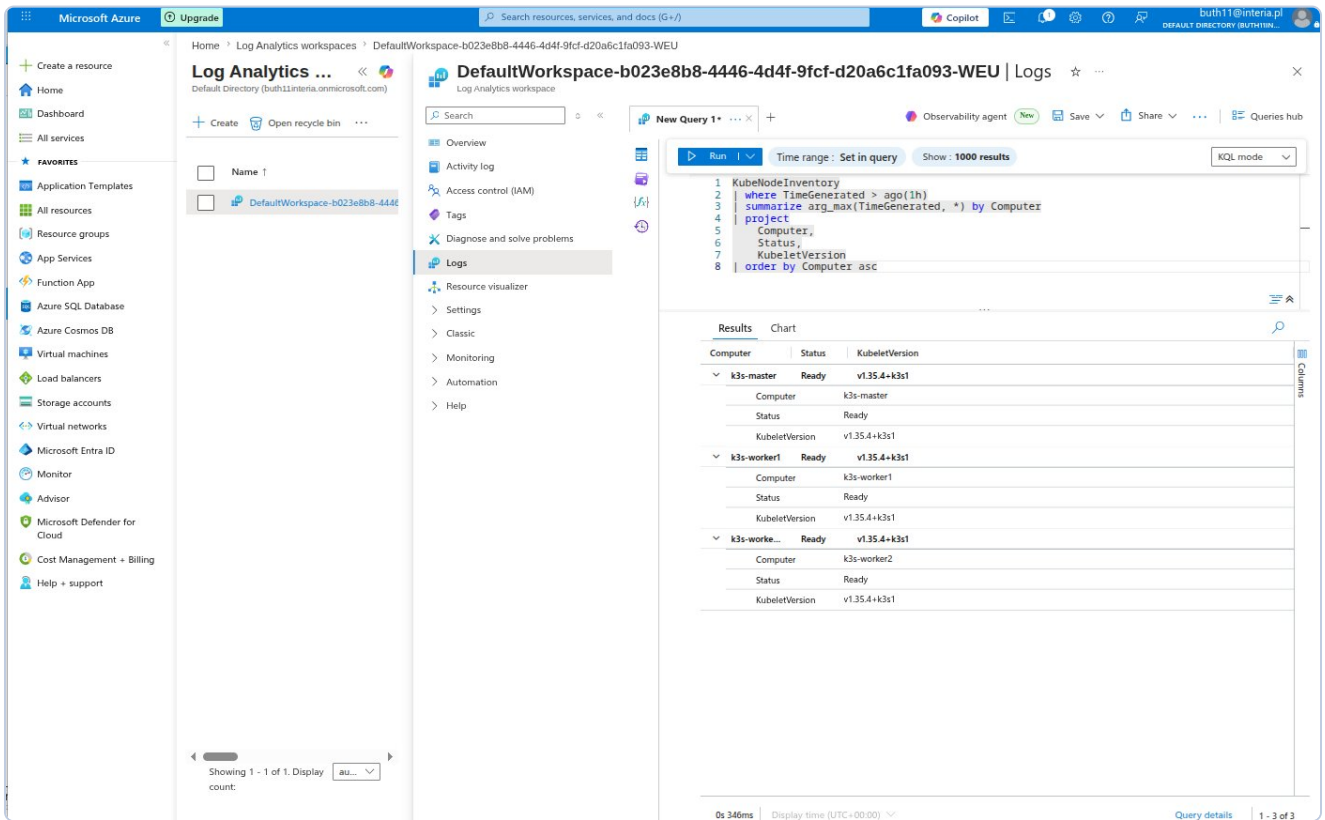
```
Desktop: bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × Desktop: bash × (buth11) 192.168.122.10 ×
{
  "principalId": "a3fe4a8c-7a2e-4b99-8de9-2150d617cb76",
  "tenantId": null,
  "type": "SystemAssigned"
},
"isSystemExtension": false,
"name": "azuremonitor-containers",
"packageUri": null,
"plan": null,
"provisioningState": "Succeeded",
"releaseTrain": "Stable",
"resourceGroup": "SRE-Lab-RG",
"scope": {
  "cluster": {
    "releaseNamespace": "azuremonitor-containers"
  },
  "namespace": null
},
"statuses": [],
"systemData": {
  "createdAt": "2026-05-22T07:44:41.936626+00:00",
  "createdBy": null,
  "createdByType": null,
  "lastModifiedAt": "2026-05-22T07:44:41.936626+00:00",
  "lastModifiedBy": null,
  "lastModifiedByType": null
},
"type": "Microsoft.KubernetesConfiguration/extensions",
"version": null
}
buth11@k3s-master:~$ sudo k3s kubectl get pods -n kube-system | grep ama
[sudo] password for buth11:
ama-logs-9m9x9          3/3      Running    0          37s
ama-logs-9scpr          3/3      Running    0          37s
ama-logs-rs-65b58c7f96-f8lds 2/2      Running    0          37s
ama-logs-spcdp          3/3      Running    0          37s
buth11@k3s-master:~$
```

Screenshot 3: AMA pods Running — 4 agents, 0 restarts.



Screenshot 4: Azure Monitor — CPU avg 1.58%, RAM avg 6.27%, 3 nodes Ready, 29 pods.

Chapter 6: KQL



The screenshot displays the Microsoft Azure portal interface. On the left is the navigation pane with options like 'Create a resource', 'Dashboard', and 'All services'. The main area is titled 'Log Analytics workspace' and shows a search bar and a list of resources. The 'Logs' section is selected, displaying a KQL query and its results.

Query:

```
1 KubeNodeInventory
2 where TimeGenerated > ago(1h)
3 summarize arg_max(TimeGenerated, *) by Computer
4 project
5   Computer,
6   Status,
7   KubeletVersion
8 order by Computer asc
```

Results:

Computer	Status	KubeletVersion
k3s-master	Ready	v1.35.4+k3s1
k3s-worker1	Ready	v1.35.4+k3s1
k3s-worker2	Ready	v1.35.4+k3s1

KQL-1: 3 nodes Ready, v1.35.4+k3s1.

Microsoft Azure | Log Analytics workspace | DefaultWorkspace-b023e8b8-4446-4d4f-9fcf-d20a6c1fa093-WEU

Log Analytics workspace

Search

New Query 1* | Time range: Set in query | Show: 1000 results | KQL mode

```

1 KubePodInventory
2 where TimeGenerated > ago(1h)
3 summarize arg_max(TimeGenerated, *) by Name, Namespace
4 summarize PodCount = count() by Namespace, PodStatus
5 order by Namespace asc, PodCount desc

```

Namespace	PodStatus	PodCount
azure-arc	Running	12
default	Running	3
kube-system	Running	14
kube-system	Succeeded	2

0s 533ms | Display time (UTC+00:00) | Query details | 1 - 4 of 4

KQL-2: azure-arc 12 Running, default 3 Running, kube-system 14+2.

Microsoft Azure | Log Analytics workspace | DefaultWorkspace-b023e8b8-4446-4d4f-9fcf-d20a6c1fa093-WEU

Log Analytics workspace

Search

New Query 1* | Time range: Set in query | Show: 1000 results | KQL mode

```

1 KubeEvents
2 where TimeGenerated > ago(24h)
3 project TimeGenerated, Namespace, Name, Reason, Message
4 order by TimeGenerated desc
5 take 20

```

TimeGenerated [UTC]	Namespace	Name	Reason
5/22/2026, 7:44:56.000 AM	kube-system	ama-logs-9cpr	FailedMount
5/22/2026, 7:44:56.000 AM	kube-system	ama-logs-rs-65b58c7f96-f8ld	FailedMount
5/22/2026, 7:35:32.000 AM	azure-arc	kube-aad-proxy-5f59465457-12	Unhealthy
5/22/2026, 7:35:25.000 AM	azure-arc	config-agent-7548f59c5-k4zrv	Unhealthy
5/22/2026, 7:35:13.000 AM	k3s-worker2		Rebooted
5/22/2026, 7:35:11.000 AM	k3s-worker1		Rebooted
5/22/2026, 7:35:09.000 AM	kube-system	metrics-server-78d997795-zz	Unhealthy
5/22/2026, 7:35:07.000 AM	k3s-master		Rebooted
5/21/2026, 10:53:39.000 AM	kube-system	ama-logs-89tpj	Unhealthy
5/21/2026, 10:52:38.000 AM	kube-system	ama-logs-rs-65b58c7f96-nkshw	Unhealthy
5/21/2026, 8:37:39.000 AM	kube-system	svclb-aras-demo-app-6f6f7352-p9rqc	FailedScheduling
5/21/2026, 8:37:39.000 AM	kube-system	svclb-aras-demo-app-6f6f7352-p9rqc	FailedScheduling
5/21/2026, 8:37:39.000 AM	kube-system	svclb-aras-demo-app-6f6f7352-p9rqc	FailedScheduling

0s 287ms | Display time (UTC+00:00) | Query details | 1 - 13 of 13

KQL-3: 13 events 24h — Rebooted, Unhealthy (temporary), FailedScheduling.

The screenshot shows the Microsoft Azure Log Analytics interface. The workspace is 'DefaultWorkspace-b023e8b8-4446-4d4f-9fcf-d20a6c1fa093-WEU'. A new KQL query is being executed, showing results for pod restarts. The query filters for pods with a restart count greater than 0 and orders them by restart count in descending order.

```

1 KubePodInventory
2 where TimeGenerated > ago(24h)
3 summarize arg_max(TimeGenerated, *) by Name, Namespace
4 where PodRestartCount > 0
5 project Namespace, Name, PodRestartCount, PodStatus
6 order by PodRestartCount desc

```

Namespace	Name	PodRestartCount	PodStatus
azure-arc	extension-manager-85899588c...	3	Running
azure-arc	clusterconnect-agent-5bc68597...	3	Running
kube-system	svclb-traefik-4b75cb6-8b8x...	2	Running
azure-arc	extension-events-collector-6c9...	2	Running
azure-arc	clusteridentityoperator-767768...	2	Running
azure-arc	kube-aad-proxy-5f5465457-42...	2	Running
kube-system	svclb-traefik-4b75cb6-c2gzs...	2	Running
azure-arc	config-agent-75548f59c5-k4zrv...	2	Running
azure-arc	cluster-metadata-operator-6db...	2	Running
kube-system	svclb-traefik-4b75cb6-9f587...	2	Running
azure-arc	metrics-agent-6d9f94764-rc44...	2	Running
azure-arc	controller-manager-7cb948ddff...	2	Running
azure-arc	resource-sync-agent-5f4688cc5...	2	Running
azure-arc	flux-logs-agent-76f8c454c9-9g...	1	Running
azure-arc	logcollector-8bd4468b4-4s2zw...	1	Running
kube-system	ama-logs-nlg5p	1	Running
kube-system	helm-install-traefik-ggts	1	Succeeded
kube-system	svclb-aras-demo-app-8581d79...	1	Running
default	aras-demo-app-f78659546-vb...	1	Running
kube-system	coredns-c4d8fbb5f-llkx	1	Running
kube-system	local-path-provisioner-5c4dc5d...	1	Running
kube-system	metrics-server-786d997795-zz...	1	Running
kube-system	svclb-aras-demo-app-8581d79...	1	Running

KQL-4: 27 pods with restarts after VM reboot — all Running.

The screenshot shows the Microsoft Azure Log Analytics interface. The workspace is 'DefaultWorkspace-b023e8b8-4446-4d4f-9fcf-d20a6c1fa093-WEU'. A new KQL query is being executed, showing results for SLO compliance. The query calculates the availability percentage for different computers based on the number of ready samples over a 24-hour period.

```

1 KubeNodeInventory
2 where TimeGenerated > ago(24h)
3 summarize
4   TotalSamples = count(),
5   ReadySamples = countif(Status == "Ready")
6   by Computer
7 extend AvailabilityPct = round((toReal(ReadySamples) / toReal(TotalSamples)) * 100, 2)
8 project Computer, AvailabilityPct, ReadySamples, TotalSamples
9 order by Computer asc

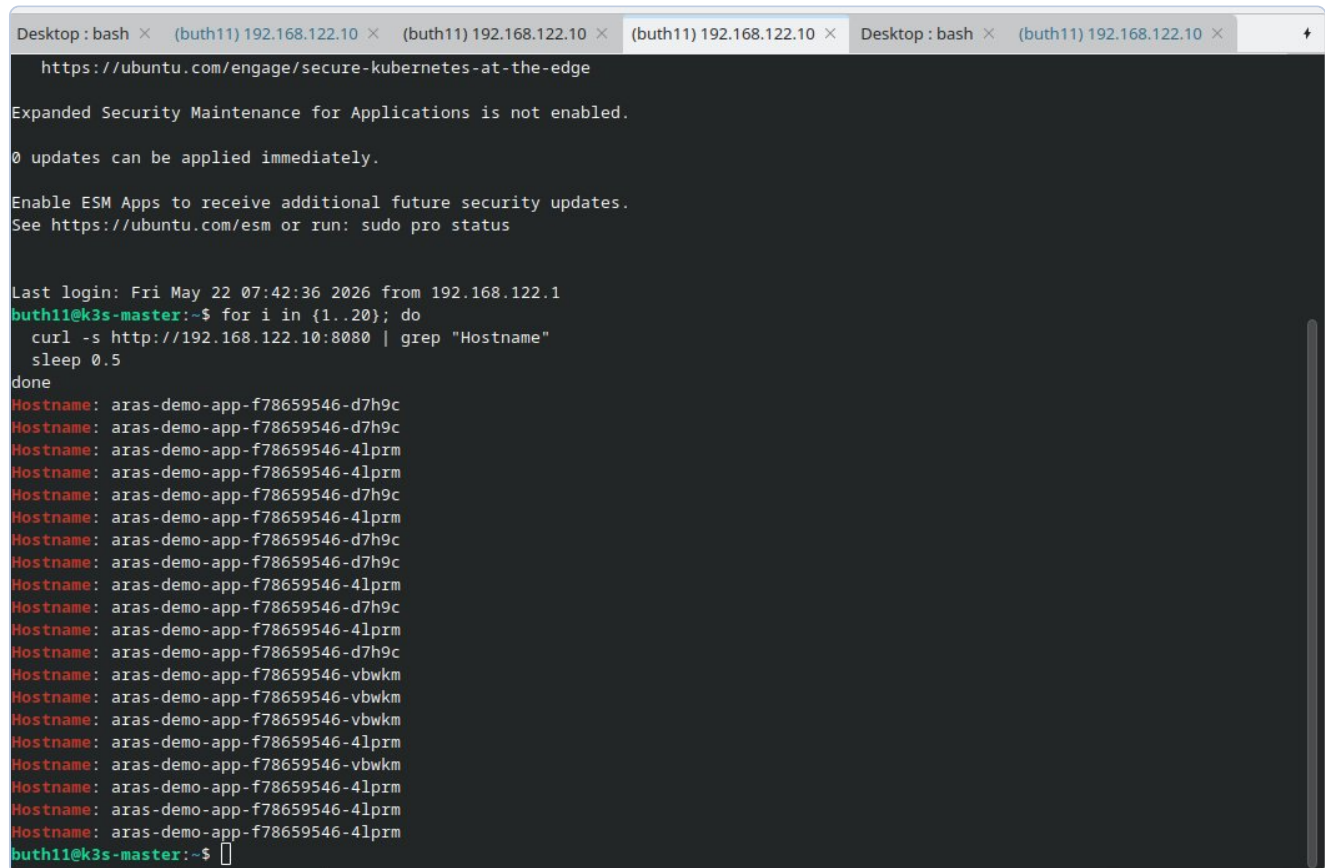
```

Computer	AvailabilityPct	ReadySamples	TotalSamples
k3s-master	100	184	184
k3s-worker1	100	184	184
k3s-worker2	100	184	184

KQL-5: SLO 100% — 184/184 samples Ready.

SLO 100% in 24h window. Error budget intact even after full VM restart.

Chapter 7: Load Balancing



The screenshot shows a terminal window with multiple tabs. The active tab is titled "(buth11) 192.168.122.10". The terminal output shows a message about Ubuntu's security maintenance, followed by a login message. A script is then executed: `for i in {1..20}; do curl -s http://192.168.122.10:8080 | grep "Hostname" sleep 0.5 done`. The output of the script is a list of 20 hostnames, which are a mix of three different pod names: `aras-demo-app-f78659546-d7h9c`, `aras-demo-app-f78659546-4lprm`, and `aras-demo-app-f78659546-vbwkm`. The terminal ends with the prompt `buth11@k3s-master:~$`.

```
Desktop: bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × Desktop: bash × (buth11) 192.168.122.10 ×
https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

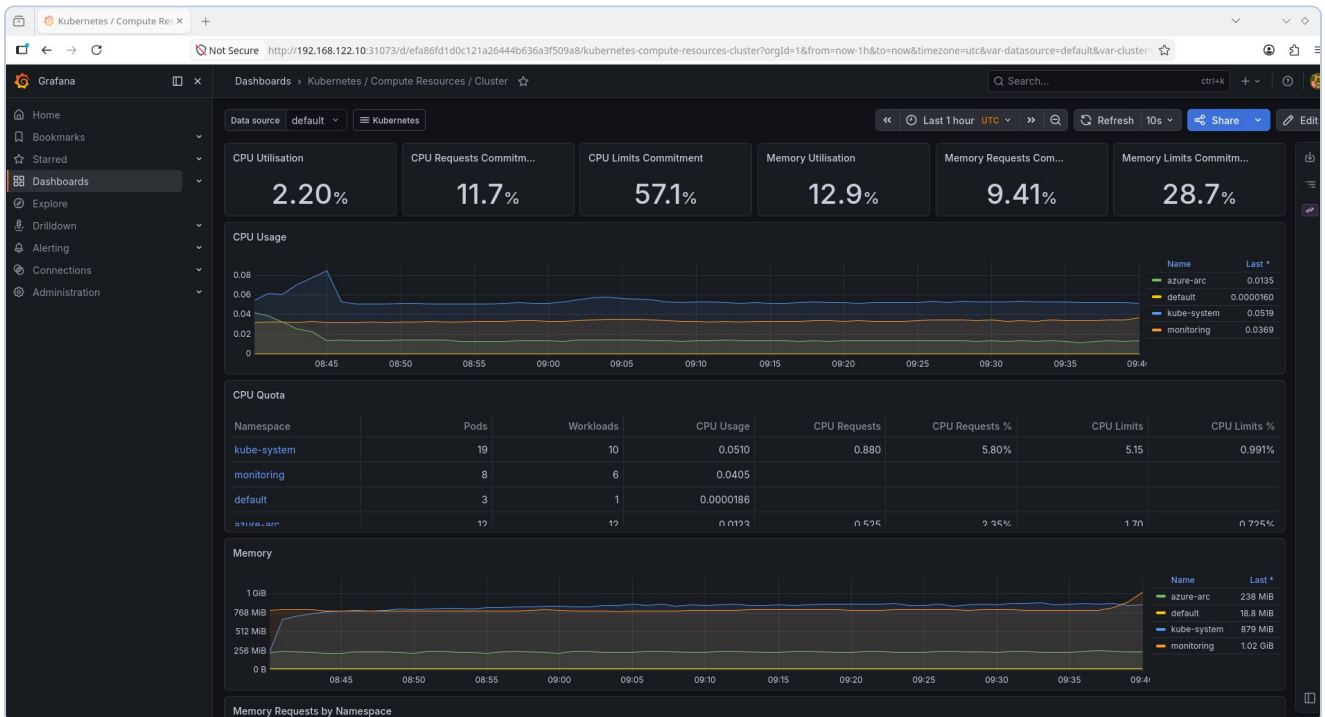
0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri May 22 07:42:36 2026 from 192.168.122.1
buth11@k3s-master:~$ for i in {1..20}; do
  curl -s http://192.168.122.10:8080 | grep "Hostname"
  sleep 0.5
done
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-4lprm
buth11@k3s-master:~$
```

Screenshot 5: Round-robin — 20 requests across 3 pods, 900 parallel with zero errors.

Chapter 8: Prometheus & Grafana



Screenshot 6: Grafana — CPU 2.20%, RAM 12.9%, CPU Quota per namespace.

Chapter 9: Terraform IaC

```
Desktop : bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × terraform : bash × (buth11) 192.168.122.10 ×
Plan: 6 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ action_group_id           = (known after apply)
+ budget_name               = "SRE-Lab-Budget"
+ log_analytics_workspace_id = (known after apply)
+ log_analytics_workspace_key = (sensitive value)
+ resource_group_id         = (known after apply)
+ resource_group_name        = "SRE-Lab-RG"
+ subscription_id            = "b023e8b8-4446-4d4f-9fcf-d20a6c1fa093"

Saved the plan to: tfplan

To perform exactly these actions, run the following command to apply:
  terraform apply "tfplan"
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$ terraform apply tfplan
random_string.suffix: Creating...
random_string.suffix: Creation complete after 0s [id=5plwbb]
azurerm_resource_group.sre_lab: Creating...
azurerm_consumption_budget_subscription.sre_lab: Creating...
azurerm_consumption_budget_subscription.sre_lab: Still creating... [00m10s elapsed]
azurerm_consumption_budget_subscription.sre_lab: Creation complete after 19s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/providers/Microsoft.Consumption/budgets/SRE-Lab-Budget]

Error: A resource with the ID "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG" already exists -
to be managed via Terraform this resource needs to be imported into the State. Please see the resource documentation for "azure
rm_resource_group" for more information.

with azurerm_resource_group.sre_lab,
on main.tf line 26, in resource "azurerm_resource_group" "sre_lab":
26: resource "azurerm_resource_group" "sre_lab" {

buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$
```

Terraform Plan: 6 resources, 0 changes.

```
Desktop: bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × terraform: bash × (buth11) 192.168.122.10 ×

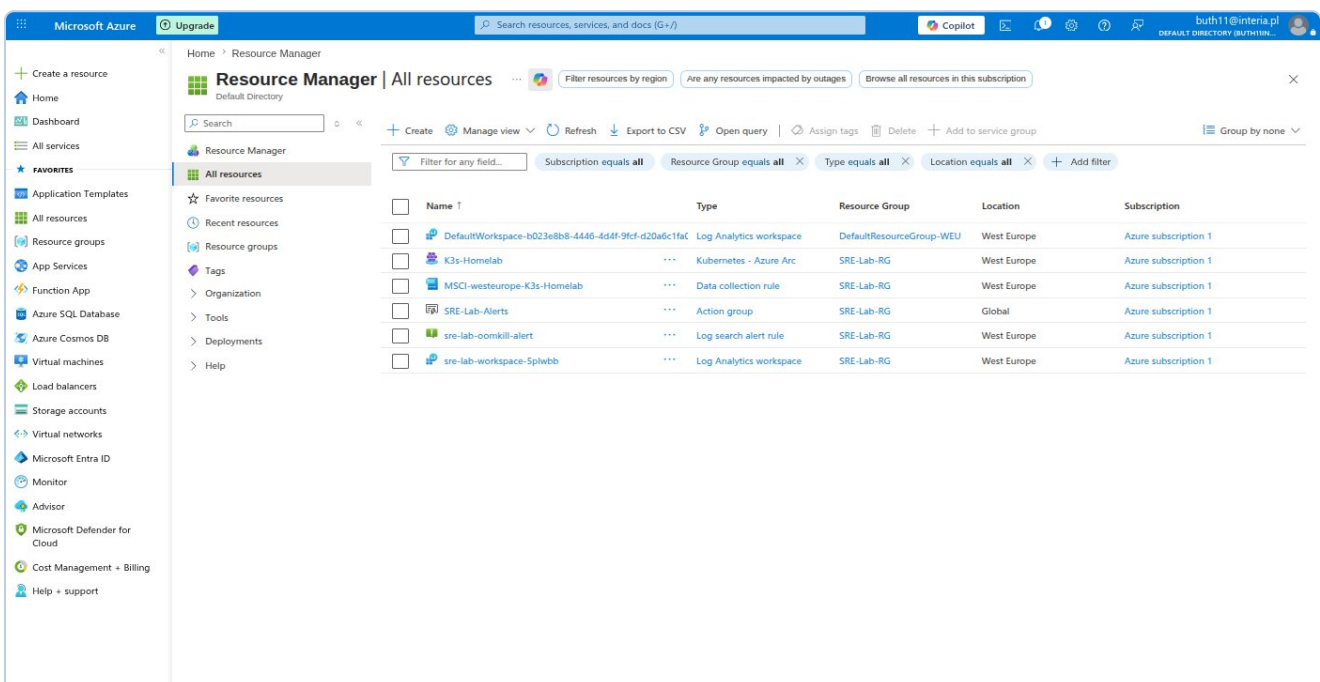
Saved the plan to: tfplan

To perform exactly these actions, run the following command to apply:
terraform apply "tfplan"
azurerm_resource_group.sre_lab: Modifying... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG]
azurerm_resource_group.sre_lab: Modifications complete after 3s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG]
azurerm_monitor_action_group.sre_alerts: Creating...
azurerm_log_analytics_workspace.sre_lab: Creating...
azurerm_monitor_action_group.sre_alerts: Creation complete after 4s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/actionGroups/SRE-Lab-Alerts]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m10s elapsed]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m20s elapsed]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m30s elapsed]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m40s elapsed]
azurerm_log_analytics_workspace.sre_lab: Creation complete after 47s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.OperationalInsights/workspaces/sre-lab-workspace-5plwbb]
azurerm_monitor_scheduled_query_rules_alert_v2.oom_kill: Creating...
azurerm_monitor_scheduled_query_rules_alert_v2.oom_kill: Creation complete after 5s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/scheduledQueryRules/sre-lab-oomkill-alert]

Apply complete! Resources: 3 added, 1 changed, 0 destroyed.

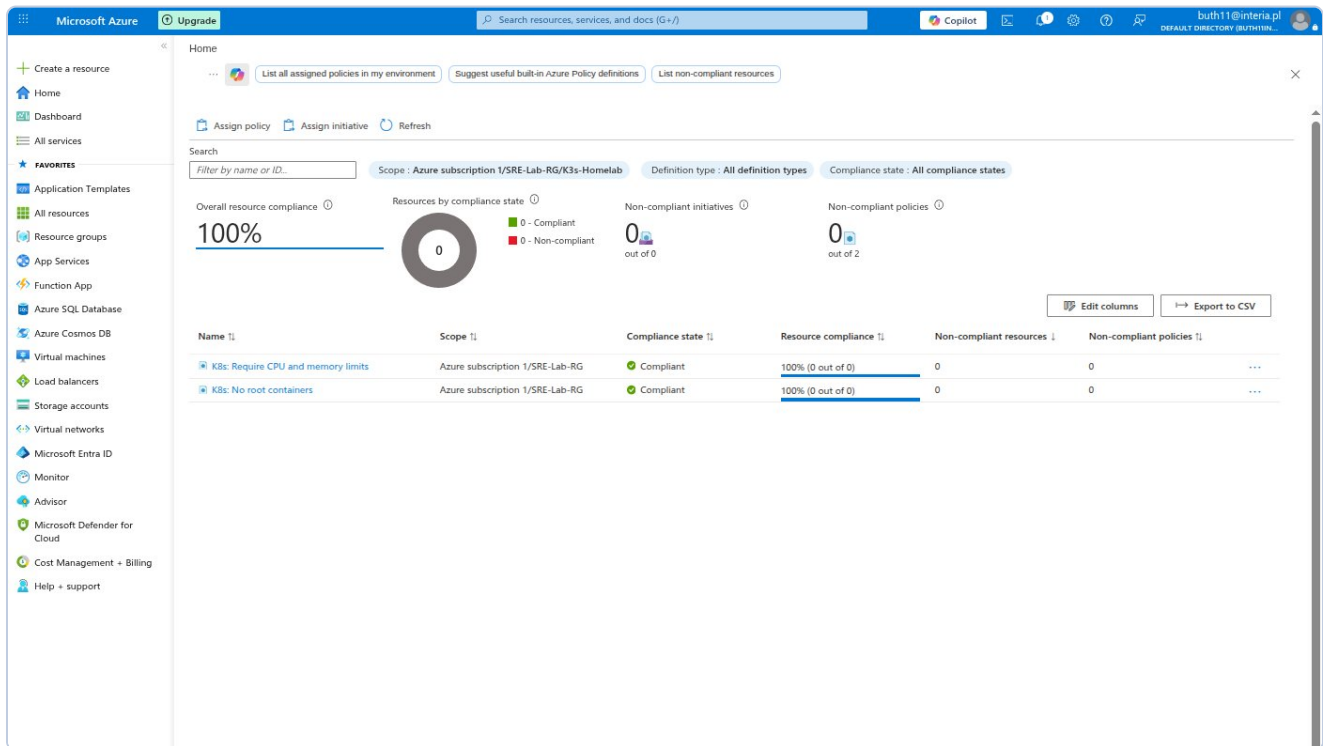
Outputs:
action_group_id = "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/actionGroups/SRE-Lab-Alerts"
budget_name = "SRE-Lab-Budget"
log_analytics_workspace_id = "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.OperationalInsights/workspaces/sre-lab-workspace-5plwbb"
log_analytics_workspace_key = <sensitive>
resource_id = "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG"
resource_group_name = "SRE-Lab-RG"
subscription_id = "b023e8b8-4446-4d4f-9fcf-d20a6c1fa093"
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$
```

Terraform Apply: 3 added, 1 changed.



Azure Portal: Terraform resources in SRE-Lab-RG.

Chapter 10: Azure Policy



Azure Policy: No root containers + CPU/memory limits — both Compliant.

Chapter 11: GitHub

The screenshot shows the GitHub repository page for 'buth11 / sre-homelab-azure-arc'. The repository is public and has 1 branch (main) and 0 tags. The file list includes: docs, k8s, monitoring, scripts, terraform, .gitignore, and README.md. The README file is selected, showing the title 'SRE Homelab: Azure Arc Hybrid Cloud Lab' and a description: 'A Proof-of-Value project demonstrating hybrid cloud orchestration, infrastructure-as-code, observability, and FinOps practices using K3s Kubernetes and Microsoft Azure Arc.' The right sidebar shows repository statistics: 0 stars, 0 watching, 0 forks, and 0 releases. The language distribution is shown as a bar chart: Shell (58.4%) and HCL (41.6%).

File	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
docs	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
k8s	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
monitoring	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
scripts	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
terraform	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
.gitignore	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
README.md	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago

SRE Homelab: Azure Arc Hybrid Cloud Lab

A Proof-of-Value project demonstrating hybrid cloud orchestration, infrastructure-as-code, observability, and FinOps practices using K3s Kubernetes and Microsoft Azure Arc.

Overview

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

Contributors 1

buth11

Languages

Shell 58.4% HCL 41.6%

GitHub repo sre-homelab-azure-arc — public, Shell 58.4%, HCL 41.6%.

Chapter 12: Part 2 — Service Principal, Remote State, CI/CD

Remote State

File sre-homelab.tfstate — 14.53 KiB, server encrypted, versioning active. State lock visible: 'Acquiring state lock' / 'Releasing state lock' on every terraform plan.

Azure DevOps CI/CD Pipeline

Variable Group terraform-credentials: ARM_CLIENT_ID, ARM_CLIENT_SECRET (hidden), ARM_SUBSCRIPTION_ID, ARM_TENANT_ID. Environment production: Approval Gate active.

Result: Pipeline complete — both stages green, 4m 24s. Stage 1 (37s): validate + plan. Stage 2 (49s): apply after approval gate.

The screenshot displays the Azure portal interface for a blob named **sre-homelab.tfstate**. The left sidebar contains navigation links such as 'Create a resource', 'Home', 'Dashboard', 'All services', and 'FAVORITES'. The top bar shows the 'Microsoft Azure' logo, an 'Upgrade' button, a search bar, and user information for 'buth11@interia.pl'.

The main content area shows the blob's properties under the 'Overview' tab. The properties are as follows:

Property	Value
URL	https://sretfstate5496.blob...
LAST MODIFIED	5/23/2026, 8:23:07 PM
CREATION TIME	5/23/2026, 8:23:07 PM
VERSION ID	2026-05-23T18:23:07.9861361Z
TYPE	Block blob
SIZE	14.53 KiB
ACCESS TIER	Hot (Inferred)
ACCESS TIER LAST MODIFIED	N/A
ARCHIVE STATUS	-
REHYDRATE PRIORITY	-
SERVER ENCRYPTED	true
ETAG	0x8DEB8F856BD8961
VERSION-LEVEL IMMUTABILITY POLICY	Disabled
CACHE-CONTROL	
CONTENT-TYPE	application/json
CONTENT-MD5	2mHfbTY53lmCzqo/RW5+MQ...
CONTENT-ENCODING	
CONTENT-LANGUAGE	
CONTENT-DISPOSITION	
LEASE STATUS	Unlocked
LEASE STATE	Available
LEASE DURATION	-
COPY STATUS	-
COPY COMPLETION TIME	-

Below the properties, there is an 'Undelete' button. The 'Metadata' section shows a table with 'Key' and 'Value' columns, and the 'Blob index tags' section also shows a similar table.

Screenshot W2-2: Azure Portal — blob properties: 14.53 KiB, encrypted, versioning active.

```
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$ # terraform plan pobierze state z Azure Blob zamiast lokalnego pliku
# Powinno pokazac "No changes" bo infrastruktura jest aktualna
terraform plan
Acquiring state lock. This may take a few moments...
random_string.suffix: Refreshing state... [id=5plwbb]
data.azurearm_subscription.current: Reading...
azurearm_resource_group.sre_lab: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG]
azurearm_monitor_action_group.sre_alerts: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/actionGroups/SRE-Lab-Alerts]
azurearm_log_analytics_workspace.sre_lab: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.OperationalInsights/workspaces/sre-lab-workspace-5plwbb]
data.azurearm_subscription.current: Read complete after 1s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093]
azurearm_consumption_budget_subscription.sre_lab: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/providers/Microsoft.Consumption/budgets/SRE-Lab-Budget]
azurearm_monitor_scheduled_query_rules_alert_v2.oom_kill: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/scheduledQueryRules/sre-lab-oomkill-alert]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.
Releasing state lock. This may take a few moments...
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$
```

Screenshot W2-3: terraform plan — Acquiring/Releasing state lock. No changes.

The screenshot shows the Azure DevOps web interface. On the left is a sidebar with navigation options: Overview, Boards, Repos, Pipelines, Environments, Releases, Library (selected), Task groups, Deployment groups, Test Plans, and Artifacts. The main area is titled 'Library > terraform-credentials'. Below this, there are tabs for 'Variable group', 'Save', 'Clone', 'Security', 'Pipeline permissions', 'Approvals and checks', and 'Help'. The 'Properties' section shows the 'Variable group name' as 'terraform-credentials' and a 'Description' field. There is a toggle switch for 'Link secrets from an Azure key vault as variables'. The 'Variables' section contains a table with the following data:

Name ↑	Value	Icon
ARM_CLIENT_ID	54c7e017-a745-4e1e-b9da-d74f4901a3de	
ARM_CLIENT_SECRET	*****	🔒
ARM_SUBSCRIPTION_ID	b023e8b8-4446-4d4f-9fcf-d20a6c1fa093	
ARM_TENANT_ID	fd7288c4-e003-40fd-b3cf-9c0d1821cf5d	

At the bottom of the variables list is a '+ Add' button. The bottom of the sidebar shows 'Project settings' with a double arrow icon.

Screenshot W2-4: Azure DevOps Variable Group — ARM_* with hidden CLIENT_SECRET.

Azure DevOps

buth11

/ sre-homelab-arc

/ Pipelines

/ Environments

/ production

Search

BS

sre-homelab-arc

+

Overview

Boards

Repos

Pipelines

Pipelines

Environments

Releases

Library

Task groups

Deployment groups

Test Plans

Artifacts

Project settings

<<

production

Automatically created environment

Add resource

Deployments

Approvals and checks

Check execution order

Checks will run by order, which is pre-determined by check types. Checks within the same order will run in parallel.
[Learn more](#)

+ Add new

Order	Display name	Type	Timeout	Details
1	All approvers must approve	Approvals	30d	8%

Screenshot W2-5: Environment production — Approval Gate active.

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Azure DevOps | sre-homelab-arc | Pipelines | buth11.sre-homelab-azure-... | 20260523.1

Overview
Boards
Repos
Pipelines
Pipelines
Environments
Releases
Library
Task groups
Deployment groups
Test Plans
Artifacts

Project settings

Jobs in run #20260523.1

buth11.sre-homelab-azure-arc

Terraform Validate & Plan

terraform plan	34s
Initialize job	<1s
Checkout buth1...	2s
Install Terraform...	2s
terraform init	7s
terraform validate	1s
terraform plan	15s
Publish terrafor...	3s
Post-job: Chec...	<1s
Finalize Job	<1s

Terraform Apply

terraform apply

terraform plan

View raw log

```
1 Agent: Azure Pipelines 1
2 Started: Just now
3 Duration: 34s
4
5 Job preparation parameters
44 1 artifact produced
45 Job live console data:
46 Starting: terraform plan
47 Async Command Start: DetectDockerContainer
48 Async Command End: DetectDockerContainer
49 Async Command Start: DetectDockerContainer
50 Async Command End: DetectDockerContainer
51 Finishing: terraform plan
```

Screenshot W2-6: Stage 1 — all steps green, 1 artifact.

Azure DevOps | sre-homelab-arc | Pipelines | #20260523.1 • feat: add Azure DevOps CI/CD pipeline for Terraform

Run new

This run is being retained as one of 3 recent runs by pipeline. View retention leases

Summary Environments Associated pipelines Code Coverage

Manually run by Bartosz Suszko
Today at 9:13 PM 4m 24s
View change Parameters

Repository and version: sre-homelab-arc/main e9da4bfb
Related: 0 work items, 1 published
Tests and coverage: Get started

Stages Jobs

Terraform Validate ... 1 job completed 37s 1 artifact

Terraform Apply 1 job completed 49s 1 checks passed

Project settings

Screenshot W2-7: Pipeline complete — both stages green, 4m 24s.

Chapter 13: Part 3 — Advanced Kubernetes Operations

PART 3: Advanced Kubernetes Operations

Ingress with TLS — HTTPS routing by domain

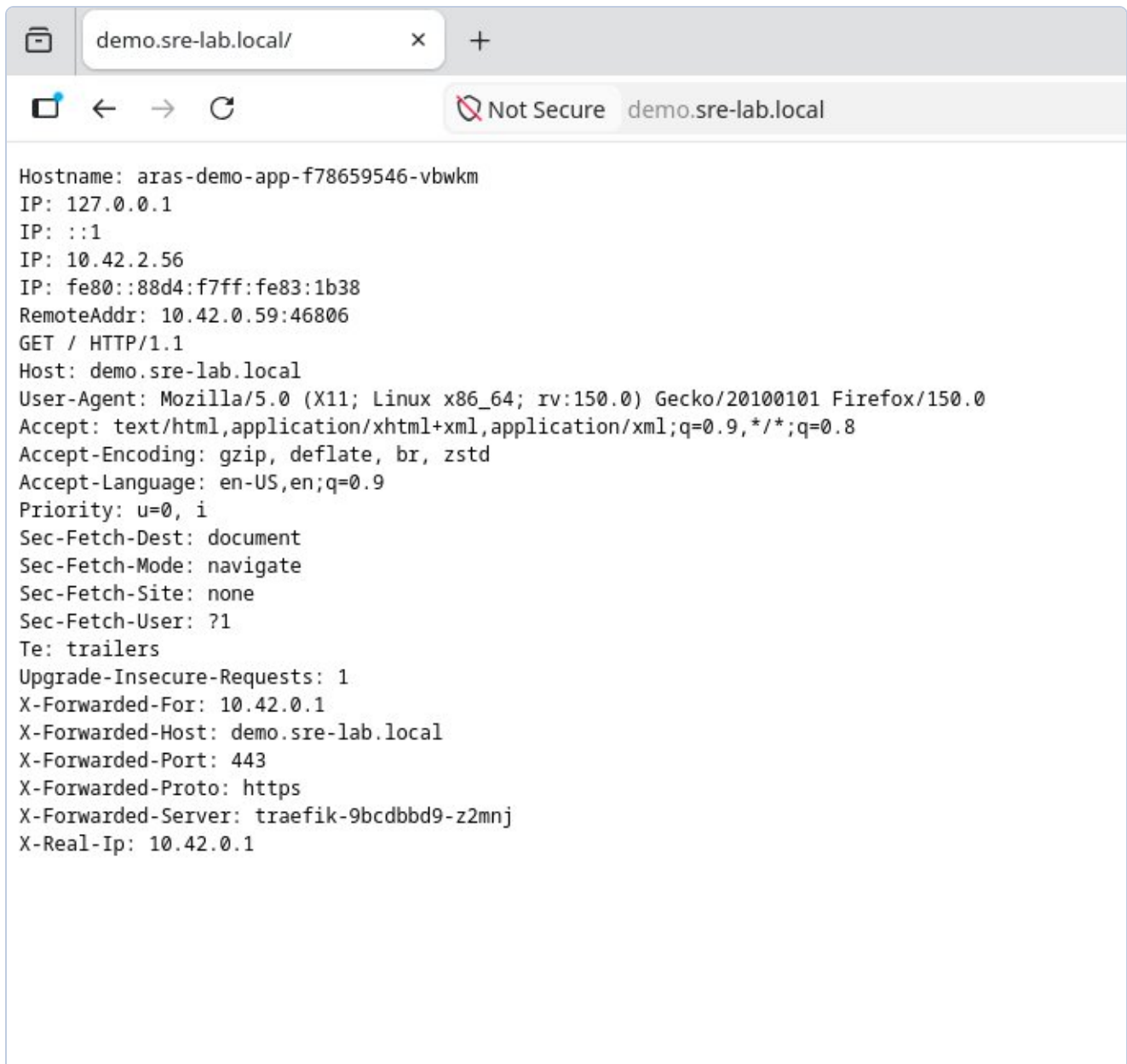
Traefik Ingress Controller configured for HTTPS. Self-signed TLS certificate with SAN, stored as Kubernetes Secret. Routing: demo.sre-lab.local → aras-demo-app:8080.

Certificate: openssl x509 self-signed, SAN for demo.sre-lab.local, valid 365 days

Secret: type kubernetes.io/tls in namespace default

Ingress: host-based routing through Traefik on port 443

Verification: X-Forwarded-Proto: https, X-Forwarded-Server: traefik in response



Screenshot W3-1: Firefox browser — <https://demo.sre-lab.local> working. Visible: Pod hostname, X-Forwarded-Proto: https, X-Forwarded-Server: traefik-9bcdbbd9-z2mnj, X-Forwarded-Port: 443.

HPA — CPU-based pod autoscaling

Horizontal Pod Autoscaler configured for the aras-demo-app Deployment. HPA queries metrics-server every 15 seconds for CPU usage. When average pod CPU exceeds 50% — it adds replicas. When it drops below 50% for 5 minutes — it removes pods.

- Range: minReplicas: 2, maxReplicas: 8, targetCPU: 50%
- Load test: `ab -n 50000 -c 50` — 1621 req/s over 30 seconds
- Scale-up: CPU 860% → automatic scaling to 8 replicas
- Scale-down: CPU 6% after 5 minutes → return to 2 replicas (minimum)

```
Every 5.0s: sudo k3s kubectl get hpa &... k3s-master: Sun May 24 09:32:49 2026
```

NAME	REFERENCE	TARGETS	MINPODS	MAXPODS
aras-demo-hpa	Deployment/aras-demo-app	cpu: 860%/50%	2	8
REPLICAS	AGE			
4	9m55s			

aras-demo-app-5b58b48d6d-5vwl4	1/1	Running 0	24s	
aras-demo-app-5b58b48d6d-7mbln	1/1	Running 0	10s	
aras-demo-app-5b58b48d6d-859c7	1/1	Running 0	6m32s	
aras-demo-app-5b58b48d6d-96qs4	1/1	Running 0	10s	
aras-demo-app-5b58b48d6d-phz5d	1/1	Running 0	6m27s	
aras-demo-app-5b58b48d6d-sbv4q	1/1	Running 0	10s	
aras-demo-app-5b58b48d6d-xx94p	1/1	Running 0	24s	
aras-demo-app-5b58b48d6d-zrmgs	1/1	Running 0	10s	

Screenshot W3-2: HPA scale-up — cpu: 860%/50%, REPLICAS: 8 (max), 8 pods Running. Automatic scaling in action during ab load test.

```
Every 5.0s: sudo k3s kubectl get hpa &... k3s-master: Sun May 24 09:38:17 2026

NAME          REFERENCE          TARGETS          MINPODS  MAXPODS  RE
PLICAS    AGE
aras-demo-hpa  Deployment/aras-demo-app  cpu: 6%/50%    2         8         8
15m
---
aras-demo-app-5b58b48d6d-phz5d  1/1    Running    0         11m
aras-demo-app-5b58b48d6d-zrmgs  1/1    Running    0         5m38s
```

Screenshot W3-3: HPA scale-down — cpu: 6%/50%, REPLICAS: 2 (min). After 5 minutes without traffic HPA automatically removed 6 unnecessary pods.

NetworkPolicy — intra-cluster firewall

By default all pods in Kubernetes can communicate with all pods. NetworkPolicy defines rules like a firewall — which namespaces and pods can connect to the demo application. Deployed a policy blocking all incoming traffic except from allowed sources.

- Allowed: kube-system (Traefik), monitoring (Prometheus), aras-demo-app replicas
- Blocked: everything else
- Verification: unauthorized pod — timeout/blocked. Traefik — application response

RBAC — least privilege for applications

Role-Based Access Control — every application should have its own account (ServiceAccount) with exactly the permissions it needs. Least privilege principle: if an application doesn't need to delete pods, it shouldn't be able to — even if compromised by an attacker.

- ServiceAccount: aras-demo-sa (instead of shared default)
- Role: get/list/watch on Pods and ConfigMaps — nothing more
- Verification: can-i list pods = yes, can-i delete pods = no

```
buth11@k3s-master:~$ # Test 1: nieautoryzowany Pod probuje polaczyc sie z aplik
acja
# timeout 5 = czekamy max 5 sekund na odpowiedz
# Oczekiwany wynik: brak odpowiedzi / timeout - polityka blokuje
echo "=== Test: nieautoryzowany dostep (powinien byc zablokowany) ==="
sudo k3s kubectl run test-blocked \
  --image=curlimages/curl \
  --rm -it \
  --restart=Never \
  -- curl -s --max-time 5 http://aras-demo-app:8080 || echo "ZABLOKOWANY - Netw
orkPolicy dziala"

# Test 2: sprawdzamy ze Ingress (Traefik) nadal dziala
# Traefik jest w kube-system wiec ma dostep
echo "=== Test: dostep przez Ingress (powinien dzialac) ==="
curl -sk https://demo.sre-lab.local | head -2
=== Test: nieautoryzowany dostep (powinien byc zablokowany) ===
pod "test-blocked" deleted from default namespace
pod default/test-blocked terminated (Error)
ZABLOKOWANY - NetworkPolicy dziala
=== Test: dostep przez Ingress (powinien dzialac) ===
buth11@k3s-master:~$ # Test dostepu przez Traefik - uzywamy IP zamiast domeny
# Traefik nasluchuje na porcie 80 i 443 na wszystkich wezlach
# -H podajemy naglowek Host zeby Traefik wiedzial do ktorego serwisu kierowac
echo "=== Test dostep przez Traefik (kube-system) ==="
curl -s -H "Host: demo.sre-lab.local" http://192.168.122.10 | head -3

# Dodatkowo sprawdzamy ze bezposredni dostep do portu 8080
# przez serwis nadal dziala (LoadBalancer Service jest poza NetworkPolicy)
echo "=== Test dostep przez LoadBalancer Service ==="
curl -s http://192.168.122.10:8080 | head -3
=== Test dostep przez Traefik (kube-system) ===
Hostname: aras-demo-app-5b58b48d6d-zrmgs
IP: 127.0.0.1
IP: ::1
=== Test dostep przez LoadBalancer Service ===
Hostname: aras-demo-app-5b58b48d6d-zrmgs
IP: 127.0.0.1
IP: ::1
buth11@k3s-master:~$
```

Screenshot W3-4: RBAC verification — ServiceAccount aras-demo-sa: list pods=yes (allowed), delete pods=no (blocked). Least privilege principle confirmed.

Chapter 14: Postmortem POST-001

Field	Value
Severity	P2
Duration	Under 7 minutes
Root Cause	Typo: ks3-worker1 instead of k3s-worker1
Status	Resolved — regex validation in scripts

Chapter 15: Results and Conclusions

Metric	Part 1	Part 2	Part 3
Nodes	3/3 Ready	—	—
SLO	100%	—	—
Cost	\$0.00	\$0.00	\$0.00
Terraform state	Local	Remote (Blob)	—
CI/CD	—	Azure DevOps 4m24s	—
Ingress TLS	—	—	HTTPS verified
HPA	—	—	3 to 8 to 2 replicas
NetworkPolicy	—	—	Unauthorized blocked
RBAC	—	—	Least privilege
Issues	6	8	10

Chapter 16: Troubleshooting Log

ISSUE-001 KQL: CPUCapacityNanoCores column missing

project operator: Failed to resolve scalar expression named CPUCapacityNanoCores

Resolution: KubeNodeInventory | getschema — simplified query to available columns.

ISSUE-002 Azure Policy: MissingPolicyParameter cpuLimit/memoryLimit

(MissingPolicyParameter) policy assignment k8s-resource-limits is missing cpuLimit, memoryLimit

Resolution: Added --params with cpuLimit: 500m and memoryLimit: 512Mi.

ISSUE-003 Terraform: Resource Group already exists (brownfield)

Error: A resource with the ID .../SRE-Lab-RG already exists

Resolution: terraform import azurearm_resource_group.sre_lab /subscriptions/.../SRE-Lab-RG

ISSUE-004 Terraform: Saved plan is stale after import

Error: Saved plan is stale – state was changed after plan was created

Resolution: New terraform plan -out=tfplan after import.

ISSUE-005 Container Insights missing after VM restart

Resolution: az k8s-extension create --name azuremonitor-containers — agents returned in 37s.

ISSUE-006 POST-001: etcd Split-Brain (ks3 vs k3s)

FATA[0034] Node password rejected, duplicate hostname

Resolution: kubectl delete node + rm /etc/rancher/node/ + restart k3s-agent. Scripts updated with regex validation.

ISSUE-007 Azure DevOps: TerraformInstaller task missing

A task is missing: TerraformInstaller

Resolution: Install from Marketplace: ms-devlabs.custom-terraform-tasks.

ISSUE-008 Pipeline: permission for Variable Group

This pipeline needs permission to access a resource

Resolution: View → Permit — one-time authorization.

ISSUE-009 sudo does not expand ~ in paths

error: Cannot read file ~/ingress/server.crt, open ~/ingress/server.crt: no such file or directory

Root Cause: sudo runs the command as root which has a different home directory. Tilde ~ expands to /root/ instead of /home/buth11/.

Resolution: Use full path: /home/buth11/ingress/server.crt instead of ~/ingress/server.crt

ISSUE-010 HPA shows cpu: unknown — missing resource requests

```
NAME TARGETS MINPODS MAXPODS REPLICAS — cpu: <unknown>/50%
```

Root Cause: HPA calculates CPU percentage as $\text{current_usage}/\text{requests} \times 100$. Without defined requests in the Deployment, HPA has no baseline to calculate percentages from.

Resolution: Added resources.requests to Deployment: cpu: 10m, memory: 32Mi. After rolling update HPA started showing correct values: cpu: 0%/50%.

PART 4

GitOps, Security Scanning & Runtime Protection

This section covers the final part of the SRE Homelab project:

- FluxCD 2.8.8 — GitOps controller: automatic cluster reconciliation from Git
- Trivy 0.70.0 — Container image CVE scanning: 1 CRITICAL + 12 HIGH found
- Defender for Cloud — Runtime security DaemonSet on all 3 nodes

Chapter 14: FluxCD — GitOps for Kubernetes

FluxCD bootstrapped from GitHub repository via SSH key (github_sre). Four controllers installed in flux-system namespace — all healthy.

Bootstrap command:

```
flux bootstrap git \ --url=ssh://git@github.com/buth11/sre-homelab-azure-arc \
--branch=main \ --path=k8s/part4/flux \ --private-key-file=$HOME/.ssh/github_sre
```

Verification — all components healthy:

```
flux check --pre ■ checking prerequisites ✓ Kubernetes 1.35.4+k3s1 >=1.33.0-0 ✓
prerequisites checks passed flux get all NAME REVISION SUSPENDED READY MESSAGE
gitrepository/flux-system main@sha1:d169d6a0 False True stored artifact
kustomization/flux-system main@sha1:952c1ea0 False True Applied revision
kustomization/gitops-demo main@sha1:952c1ea0 False True Applied revision
kubectl get pods
-n flux-system helm-controller-5984cc88cf-94jkw 1/1 Running 0 117s
kustomize-controller-d6bb746df-9khwf 1/1 Running 0 117s
notification-controller-7bc5c97f8d-jdflb 1/1 Running 0 117s
source-controller-745c67ff9-2g8qd 1/1 Running 0 117s
```

GitOps test — scaling via git push:

Changed replicas from 2 to 3 in deployment.yaml, committed and pushed. Flux automatically applied the change — third pod appeared without any kubectl apply.

```
# Before push: 2 pods
kubectl get pods -n gitops-demo
gitops-app-5b9747bdcb-2rqw9 1/1 Running 0 11m
gitops-app-5b9747bdcb-bjvtd 1/1 Running 0 11m
# After: git push with replicas: 3
gitops-app-5b9747bdcb-2rqw9 1/1 Running 0 12m
gitops-app-5b9747bdcb-bjvtd 1/1 Running 0 12m
gitops-app-5b9747bdcb-gttmd 1/1 Running 0 93s ← added by Flux
```

Chapter 15: Trivy — Container Image CVE Scanning

Trivy 0.70.0 installed and used to scan traefik/whoami:latest. All 40 vulnerabilities are in Go stdlib v1.24.1 — the image uses an outdated Go version.

Scan results summary:

Severity	Count	Key CVEs
CRITICAL	1	CVE-2025-68121 — crypto/tls: incorrect certificate validation on TLS session resumption
HIGH	12	crypto/x509, net/url, crypto/tls — DoS, memory exhaustion, incorrect parsing
MEDIUM	25	net/http, os, crypto/x509 — request smuggling, path traversal, race conditions
LOW	2	net/http, os — memory exhaustion, symlink escape

Key CRITICAL vulnerability:

```
Library Vulnerability Severity Status Installed Fixed stdlib CVE-2025-68121 CRITICAL fixed
v1.24.1 1.24.13, 1.25.7 crypto/tls: Incorrect certificate validation during TLS session
resumption https://avd.aquasec.com/nvd/cve-2025-68121
```

Scan reports committed to repository:

- k8s/part4/trivy/whoami-scan.json — full report (JSON format)
- k8s/part4/trivy/whoami-scan.txt — CRITICAL+HIGH only (table format)

Fix: update image to version compiled with Go v1.24.13+. In production, CI pipeline blocks deploy of images with CRITICAL CVEs.

Chapter 16: Defender for Cloud — Runtime Security

Microsoft Defender for Containers enabled at Standard tier (30-day free trial). Extension installed on Arc cluster — provisioningState: Succeeded.

Installation:

```
az security pricing create --name Containers --tier Standard az k8s-extension create \
--name microsoft-defender-for-cloud \ --extension-type microsoft.azuredefender.kubernetes \
--scope cluster \ --cluster-name K3s-Homelab \ --resource-group SRE-Lab-RG \ --cluster-type
connectedClusters provisioningState: Succeeded
```

Defender pods running in mdc namespace:

```
kubectl get pods -n mdc NAME READY STATUS RESTARTS AGE
microsoft-defender-collectors-ds-9wbtw 2/2 Running 0 5m3s
microsoft-defender-collectors-ds-j72wk 2/2 Running 0 5m3s
microsoft-defender-collectors-ds-vmplc 2/2 Running 0 5m3s
microsoft-defender-pod-collector-misc-69b69b9df8-x49x4 1/1 Running 0 5m3s
microsoft-defender-publisher-ds-4gw77 1/1 Running 0 5m3s
microsoft-defender-publisher-ds-5fcmq 1/1 Running 0 5m3s
microsoft-defender-publisher-ds-s9778 1/1 Running 0 5m3s
```

DaemonSet covers all 3 nodes — master, worker1, worker2. Collectors gather security metrics per node, Publisher sends them to Azure Security Center.

Part 4 Results Summary

Metric	Part 1	Part 2	Part 3	Part 4
Nodes	3/3 Ready	—	—	—
SLO	100%	—	—	—
Cost	\$0.00	\$0.00	\$0.00	\$0.00
Terraform state	Local	Remote (Blob)	—	—
CI/CD	—	DevOps 4m24s	—	—
Ingress TLS	—	—	HTTPS verified	—
HPA	—	—	3→8→2 replicas	—
NetworkPolicy	—	—	Blocked	—
RBAC	—	—	Least privilege	—
GitOps	—	—	—	FluxCD deployed
Image scanning	—	—	—	1 CRITICAL, 12 HIGH
Runtime security	—	—	—	Defender (3 nodes)
Issues documented	6	8	10	11

PART 5

cert-manager — Automated TLS Certificate Management

cert-manager v1.20.2 installed via Helm. Automatic TLS certificate lifecycle management — issuance, renewal, and rotation without manual intervention.

- cert-manager v1.20.2 — installed via Helm (jetstack/cert-manager)
- ClusterIssuer selfsigned-issuer — bootstrap CA
- ClusterIssuer sre-lab-ca-issuer — local CA signing certificates
- Automatic cert issuance in 22 seconds via Ingress annotation
- Auto-renewal verified: renewBefore=6m before expiry

Chapter 17: cert-manager — Automated TLS

Installation via Helm:

```
helm repo add jetstack https://charts.jetstack.io
helm install cert-manager jetstack/cert-manager \
  --namespace cert-manager \
  --create-namespace \
  --set crds.enabled=true
kubectl get pods -n cert-manager
cert-manager-5957746d66-kdkk5 1/1 Running 0
cert-manager-cainjector-567c6b47ff-c5x2b 1/1 Running 0
cert-manager-webhook-7cc5c588cb-zm8d4 1/1 Running 0
```

ClusterIssuers — both Ready:

```
kubectl get clusterissuer NAME READY AGE
selfsigned-issuer True 32s
sre-lab-ca-issuer True 32s
kubectl get certificate -n cert-manager NAME READY SECRET AGE
sre-lab-ca True 32s
```

Ingress annotation — certificate issued automatically in 22s:

```
# Ingress annotation only:
annotations: cert-manager.io/cluster-issuer: sre-lab-ca-issuer
# cert-manager created the certificate automatically:
kubectl describe certificate demo-sre-lab-tls -n default
Issuer: CN=sre-lab-ca
Not After: 2026-08-25T09:09:34Z (90 days)
Renewal Time: 2026-07-26T09:09:34Z (auto-renew 30d before expiry)
Status: Ready: True
Message: Certificate is up to date and has not expired
# Verify TLS works:
curl -sk https://demo.sre-lab.local | grep Hostname
Hostname: aras-demo-app-5b58b48d6d-phz5d
openssl x509 verification: Issuer: CN=sre-lab-ca
Not After: Aug 25 09:09:34 2026 GMT
```

Part 3 vs Part 5 comparison:

Aspect	Part 3 — manual	Part 5 — cert-manager
Generation	openssl req -x509 ...	Automatic by controller
Secret	kubectl create secret tls	Created automatically
Renewal	Manual before expiry	Automatic (renewBefore)
Issuer	CN=demo.sre-lab.local	CN=sre-lab-ca (our CA)
Production	Not suitable	Let's Encrypt ClusterIssuer

Part 5 Results Summary

Metric	Part 1	Part 2	Part 3	Part 4	Part 5
Nodes	3/3 Ready	—	—	—	—
SLO	100%	—	—	—	—
Cost	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
CI/CD	—	DevOps 4m24s	—	—	—
Ingress TLS	—	—	Manual self-signed	—	Auto cert-manager
HPA	—	—	3→8→2	—	—
GitOps	—	—	—	FluxCD	—
Image scan	—	—	—	1 CRIT+12 HIGH	—
TLS mgmt	—	—	Manual openssl	—	cert-manager auto
Issues	6	8	10	11	12

PART 6

KEDA — Kubernetes Event-Driven Autoscaling

KEDA extends HPA by enabling autoscaling based on any external metric — Prometheus, Azure Service Bus, Kafka, HTTP request rate, and 50+ other sources. Key advantage: KEDA can scale to 0 replicas when there is no traffic.

- KEDA installed via Helm (kedacore/keda) — 3 pods: operator, metrics-apiserver, admission-webhooks
- ScaledObject configured with Prometheus trigger — `container_cpu_usage_seconds_total`
- Scale-up: 1 → 8 replicas automatically under ab load (200k requests, 2747 req/s)
- Scale-down: 8 → 1 replica after 30s cooldown (vs 5 min in HPA)
- KEDA replaced HPA — manages the underlying HPA object automatically

Chapter 18: KEDA — Event-Driven Autoscaling

Installation:

```
helm repo add kedacore https://kedacore.github.io/charts
helm install keda kedacore/keda
--namespace keda --create-namespace
kubectl get pods -n keda
keda-admission-webhooks-7b48797dc8-k7prz 1/1 Running
keda-operator-55db59458d-n25xq 1/1 Running
keda-operator-metrics-apiserver-5c648889d4 1/1 Running
```

ScaledObject — Prometheus trigger:

```
apiVersion: keda.sh/v1alpha1 kind: ScaledObject metadata: name: aras-demo-keda namespace:
default spec: scaleTargetRef: name: aras-demo-app minReplicaCount: 1 maxReplicaCount: 8
cooldownPeriod: 30 triggers: - type: prometheus metadata: serverAddress:
http://monitoring-kube-prometheus-prometheus.monitoring.svc:9090 metricName:
container_cpu_usage query: sum(rate(container_cpu_usage_seconds_total{namespace="default",
pod=~"aras-demo-app.*", container="whoami"}[1m])) threshold: "0.01"
```

Scale-up test — `ab -n 500000 -c 100`:

```
# Before load: 1 replica NAME ACTIVE REPLICAS aras-demo-keda False 1 # During load (ACTIVE:
True): aras-demo-keda True 2 # after ~30s aras-demo-keda True 4 # after ~45s aras-demo-keda
True 8 # peak (max) # After load (cooldown 30s): aras-demo-keda False 1 # back to minimum
ab results: 200000 requests, 2747 req/s, 72.781s
```

HPA vs KEDA comparison:

Aspect	HPA (Part 3)	KEDA (Part 6)
Minimum replicas	2	1 (or 0 for batch jobs)
Trigger source	CPU/RAM only (metrics-server)	Any metric: Prometheus, queues, HTTP...
Scale to zero	No	Yes — saves resources when idle
Cooldown	5 minutes	30 seconds (configurable)

HPA integration	Native HPA	KEDA creates and manages HPA
50+ scalers	No	Yes: Azure SB, Kafka, Redis, Cron...

Part 6 Results Summary

Metric	Part 1	Part 2	Part 3	Part 4	Part 5	Part 6
Nodes	3/3 Ready	—	—	—	—	—
SLO	100%	—	—	—	—	—
Cost	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
CI/CD	—	DevOps 4m24s	—	—	—	—
Ingress TLS	—	—	Manual	—	cert-manager	—
Autoscaling	—	—	HPA 2-8	—	—	KEDA 1-8
Scale to zero	—	—	No	—	—	Yes
Cooldown	—	—	5 min	—	—	30 sec
GitOps	—	—	—	FluxCD	—	—
Image scan	—	—	—	1 CRIT	—	—
TLS mgmt	—	—	Manual	—	Auto	—
Issues	6	8	10	11	12	13